

Supplementary document to ‘A covarion model for phylogenetic estimation using discrete morphological datasets’

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This document is a supplement accompanying the manuscript ‘A covarion model for phylogenetic estimation using discrete morphological datasets’. It consists of the data sources used in the main manuscript for testing the *Covariomorph* model. In addition to that this document also includes a few introductory figures about the model (Figs. [S1](#), [S3](#), [S5](#), and [S4](#)), coverage plots validating the implementation in RevBayes (Fig. [S2](#)), and some results (Figs. [S7](#), [S9](#), [S8](#), [S10](#), [S11](#)).

Graphical representation of the Covariomorph model

We can visualize the Covariomorph model by following a graphical model style ([Jordan, 2004](#)) introduced in phylogenetics by [Höhna et al. \(2014\)](#). Figure [S1](#) illustrates the graphical model representation of the covariomorph model used in this study. The phylogenetic tree (ψ) is defined by a uniformly distributed topology τ over N taxa and branch lengths $bl_i \sim \text{Exponential}(\lambda)$, where $i \in 2N - 3$ and λ is 10. Morphological character evolution is modeled using a phylogenetic continuous-time Markov chain (PhyloCTMC) parameterized by ψ and a covariomorph rate matrix Q_{cov} . The covariomorph model introduces heterogeneity in character transition rates via a set of j rate categories, each associated with a rate multiplier $m_j \sim \text{LogNormal}(\mu_m = 0, \sigma_m)$, where σ is a hyperparameter controlling the magnitude of rate variation. Switching between the rate categories occur at a uniform rate ($\delta \sim \text{Uniform}$), and within each rate categories, character transitions follow a standard Mk model (Q_{Mk}). The combined process Q_{cov} captures both among-character and temporal rate heterogeneity by integrating over the hidden rate process, and is used to generate morphological data or infer the phylogeny from the morphological data *seq*.

Validation of Implementation

Validating the implementation of the covariomorph model in RevBayes was done by Simulation Based Calibration ([Talts et al., 2018](#)). According to this approach, if we draw parameter values from their prior distribution and subsequently simulate data using these parameter values, the inferred credible intervals will contain the true parameter value in the same percent of the repetitions. In our test, we used a ‘true’

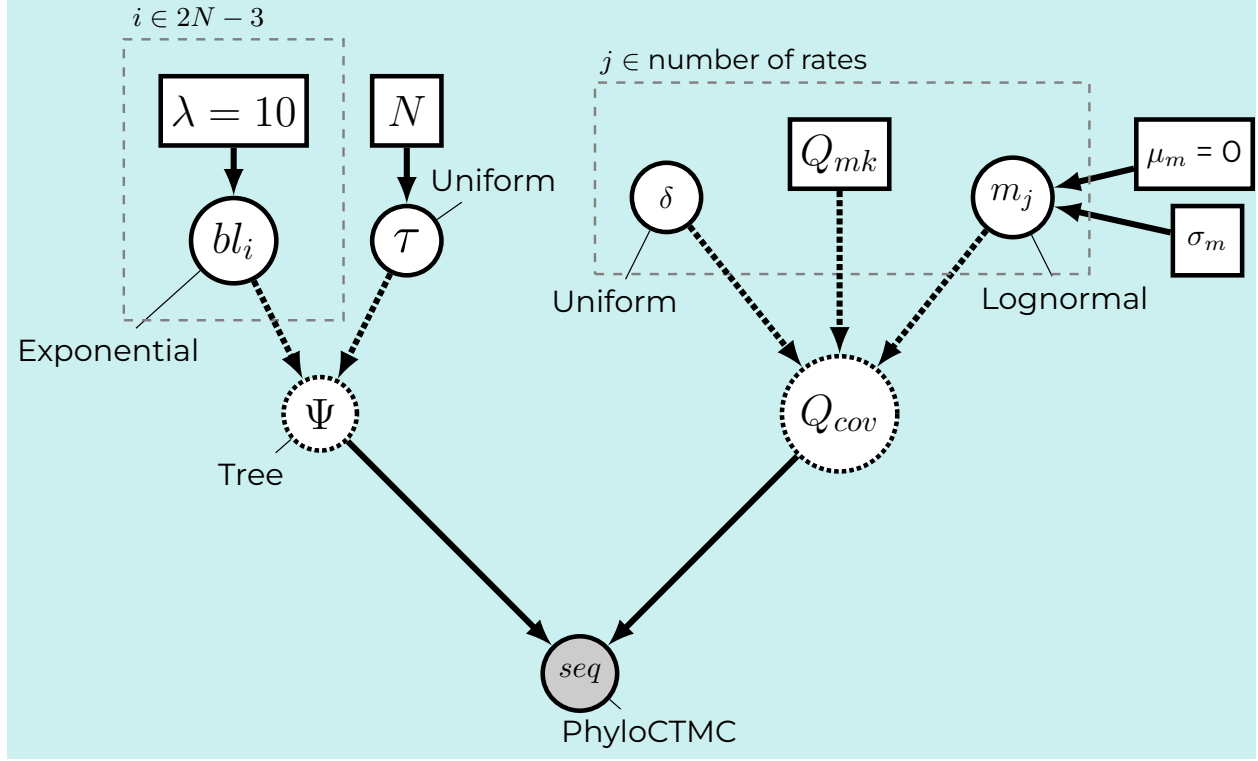


Figure S1: A graphical model displaying the components of the covariomorph model setup in the study. Following [Höhna et al. \(2014\)](#).

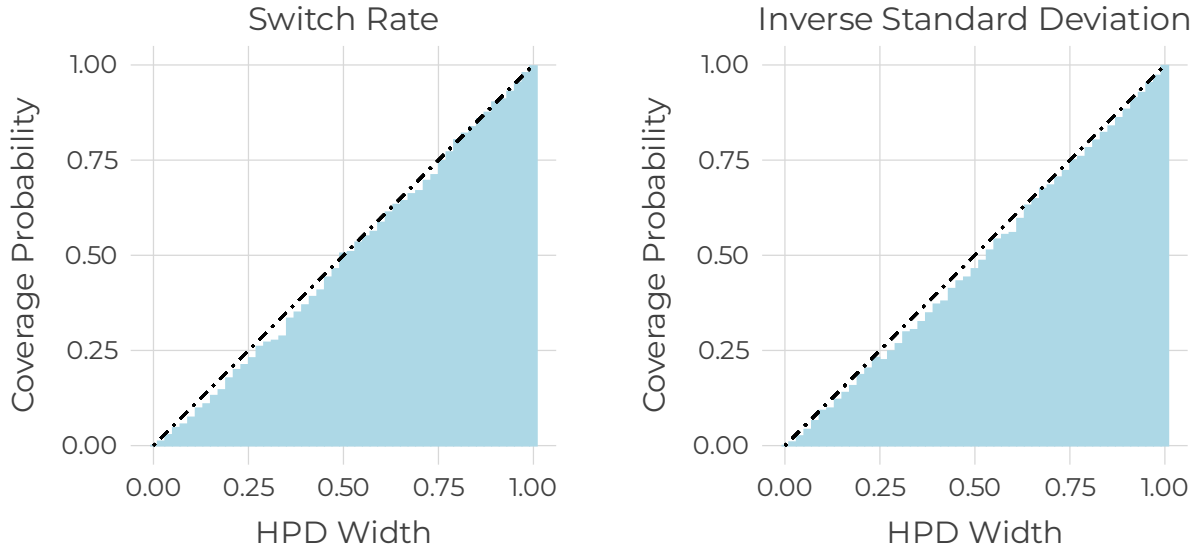


Figure S2: Results of the validation analysis for the covarion model implementation. These plots show how well the highest posterior density (HPD) captures the true parameter values across different interval widths.

value for the inverse of standard deviation of the lognormal distribution and a ‘true’ value for the switching rate to generate datasets. Upon performing MCMC, the estimated value for the standard deviation should

fall within the 95% HPD (highest posterior density) at least 90% of the time for the coverage to be 0.9. The results from our validation analyses is visualized in Figure S2.

Virtual states expansion

In the Covariomorph model, the observed character states at the tips of the phylogeny are expanded into “virtual states” to account for different evolutionary rate regimes. Figure S3 illustrates this process for a character with two observed states (0 and 1) and three rate categories. Each observed state is expanded into a set of virtual states, where each virtual state corresponds to the observed state evolving under a specific rate category. For instance, an observed state of ‘1’ is expanded into the set of virtual states $\{1, 3, 5\}$, and an observed state of ‘0’ becomes $\{0, 2, 4\}$. This expansion allows the model to simultaneously infer the character state and the evolutionary rate category at any point in the tree, capturing the heterogeneity of the evolutionary process.



Figure S3: Virtual states obtained from the observed states when the number of rate category is 3.

Computational Requirements

The covariomorph model necessarily introduces additional computational costs compared to the standard Mk model. This increase is driven by two factors:

1. **Additional Parameters:** The model adds two new global parameters to be estimated: the shape parameter of the rate distribution (σ in our case) and the switching rate (δ). This is one more parameter than the Mk+ACRV model.
2. **Expanded State Space:** The primary increase in computational load comes from the expansion of the instantaneous rate matrix (Q-matrix). The model uses m hidden rate categories, expanding the k -state character matrix to a $k \times m$ “virtual state” matrix. All likelihood calculations are performed on this larger matrix.

Runtime Analysis

We performed a benchmark analysis to quantify the relative runtime of the covariomorph model compared to the Mk and Mk+ACRV models.

Methods:

- We simulated datasets varying in both the number of taxa (10, 20, 30) and the number of characters (50, 100, 200, 300, 400).
- We compared the performance of five model configurations: Mk, Mk+ACRV ($m = 4$), Mk+ACRV ($m = 8$), Covariomorph ($m = 4$), and Covariomorph ($m = 8$).
- All benchmark analyses were performed on a single compute node CPU (AMP EPYC 9654 2.4 Ghz) using RevBayes compiled without parallelization.
- Each run consisted of a single MCMC chain for 50,000 generations.

Results:

The results of the benchmark are presented in Figure S4.

- The covariomorph model is the most computationally demanding, and its runtime scales significantly with the number of rate categories (m).
- With 8 rate categories, the covariomorph model can be over two orders of magnitude slower than the baseline Mk model, particularly on datasets with fewer taxa where the relative cost of the complex Q-matrix calculation is the highest.
- The Mk+ACRV model shows a similar, though less pronounced, increase in cost as m increases.
- While the covariomorph model is computationally intensive, it remains a practical and feasible option for the dataset sizes typically used in morphological phylogenetics.

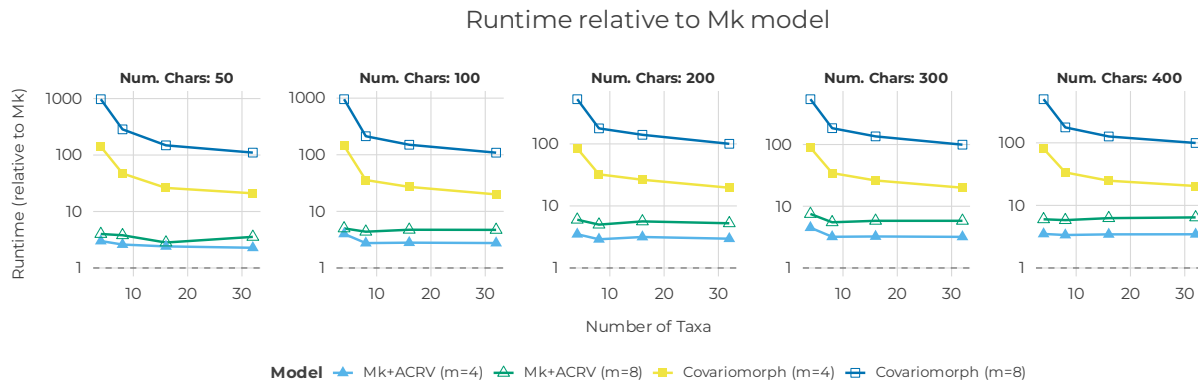


Figure S4: Relative runtime of Mk, Mk+ACRV and Covariomorph models across datasets varying in the number of taxa and characters compared to the baseline Mk model. Values greater than one indicate slower performance relative to Mk. The covariomorph model is consistently more demanding than Mk or Mk+ACRV, especially with higher number of rate categories, but remains practical for datasets of typical size.

Memory Requirements

The memory usage for likelihood calculation also scales with the model's complexity. Following [Smith et al. \(2024\)](#), the memory required grows linearly with the number of taxa, the number of site patterns, the number of character states (k), and crucially for this model, the number of rate categories (m).

This linear scaling is a direct result of the expanded $k \times m$ virtual state space used for the likelihood calculations. For all the datasets analyzed in this study, the memory demands remained well within the limits of a typical modern workstation.

Discretization of the Lognormal distribution

The covariomorph model utilizes a discretized lognormal distribution to model variation in evolutionary rates across different branches and characters. Figure S5 provides a visual representation of how the rate categories are derived from this distribution. This figure illustrates the relationship between the number of discrete rate categories k and the shape parameter σ of the lognormal distribution. The plots demonstrate two key points:

- As the shape parameter σ (x-axis) increases, the variance among the rate categories also increases, leading to a wider spread between the fastest and slowest rates.
- As the number of rate categories k (different facets) increases, the continuous lognormal distribution is approximated more finely.

But it comes with a caveat. A notable consequence of using many rate categories (e.g., $k \geq 10$) is that a substantial portion of these categories are assigned very low rate values (≤ 0.1). Since each category is equiprobable, this can lead to an unintentional up-weighting of slower rate categories.

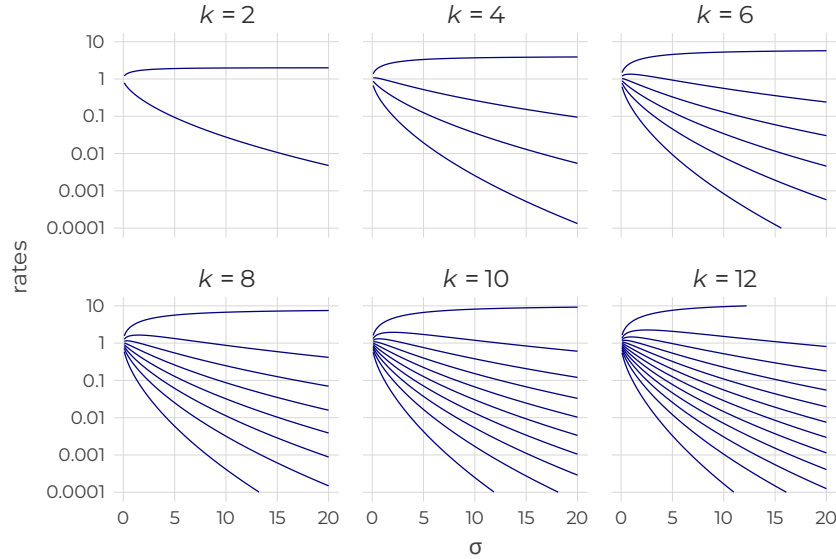


Figure S5: Rate categories from a discretized lognormal distribution. Each panel shows the values of the discrete rate categories as a function of the distribution's shape parameter, σ , for a different number of total categories (k). Within each model, the category rates (y-axis) are chosen to have equal probability ($p = 1/k$).

Topological Difference in Simulated Datasets

To evaluate the impact of model misspecification on tree topology, we conducted comparison test on simulated datasets. Using `convenience` (Fabreti and Höhna, 2022), we performed the expected difference in split frequencies test similar to how convergence is assessed. We specifically chose dataset generated under three distinct true models:

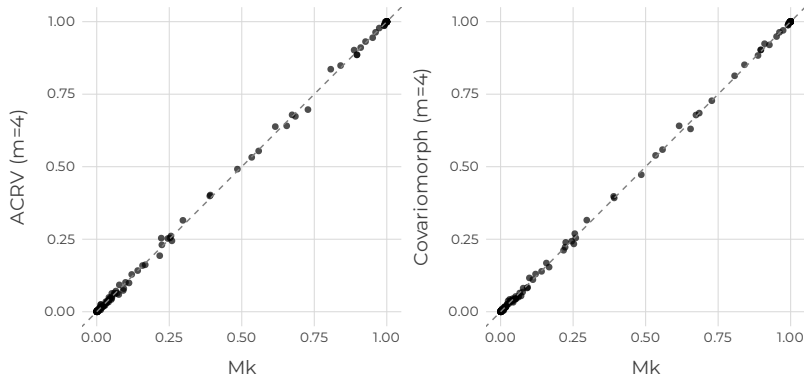
1. Mk model ($\sigma = 0$, $\delta = 0$): Data generated with simple time-homogeneous rates.
2. Mk + ACRV model ($\sigma = 1.174$, $\delta = 0$): Data generated with static rate variation across characters.
3. Covariomorph model ($\sigma = 1.174$, $\delta = 0.5$): Data generated with dynamic rate variation and switching between the rate categories.

Each of these datasets were independently analyzed under all three inference models (Mk, MK+ACRV, and Covariomorph) using the same MCMC settings described in the main manuscript.

We assessed the similarity of the posterior tree distribution inferred by each model using the expected difference in split frequencies test. This test determines if the posterior probability calculated for a given clade (split) differs significantly between two models, beyond what is expected from Monte Carlo sampling error. Clades falling outside the threshold (defined by an Effective Sample Size, $ESS = 200$ in our test) are considered to have significantly different posterior probabilities, indicating that the two models explored different regions of the tree topology space.

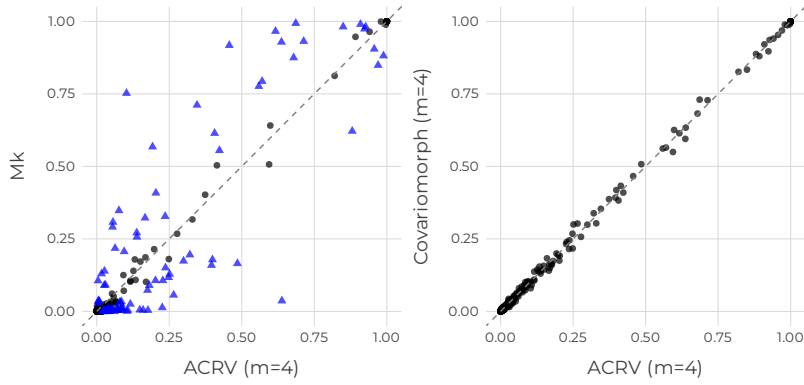
Figure S6 demonstrates the posterior probability of each unique clade observed in the joint posterior distribution for two compared models. Clades with significantly different posterior probabilities are highlighted, providing a direct visual representation of topological discordance under model misspecification.

Data Simulated using Mk



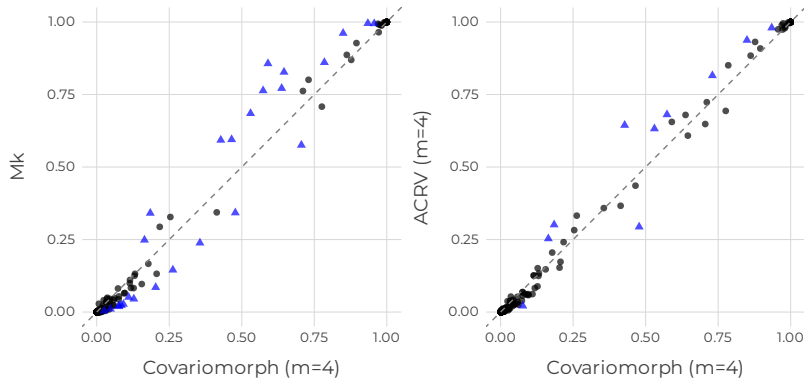
(a) Clade posterior probabilities when the true model is Mk

Data Simulated using ACRV ($m=4$)



(b) Clade posterior probabilities when the true model is Mk + ACRV ($m = 4$)

Data Simulated using Covariomorph ($m=4$)



(c) Clade posterior probabilities when the true model is Covariomorph ($m = 4$)

Figure S6: Clade posterior probabilities for different true models using simulated datasets.

Empirical datasets: Size and parameter estimates

Figure S7 displays the landscape of the empirical morphological datasets used in this study, plotting the number of taxa against the number of characters for each dataset. The datasets vary considerably in size, ranging from just a few taxa and characters to over 100 taxa and hundreds of characters.

The plots are colored according to the values of key parameters inferred by the covariomorph model with four rate categories:

- Standard Deviation, σ (Panel a): This plot shows the inferred standard deviation of the lognormal distribution that indicates the magnitude of rate heterogeneity. There is a slight correlation between the dataset size and the inferred value of σ , with low values appearing mostly for datasets with lower number of taxa and characters.
- Switching Rate, δ (Panel b): This plot illustrates the inferred rate of switching between different evolutionary rate categories. A potential trend can be observed where datasets with fewer characters and taxa (yellow points) are better explained by simpler model, such as Mk. This suggests that a larger number of characters may be necessary to provide sufficient information for the model to infer meaningful rate switching across the phylogeny.

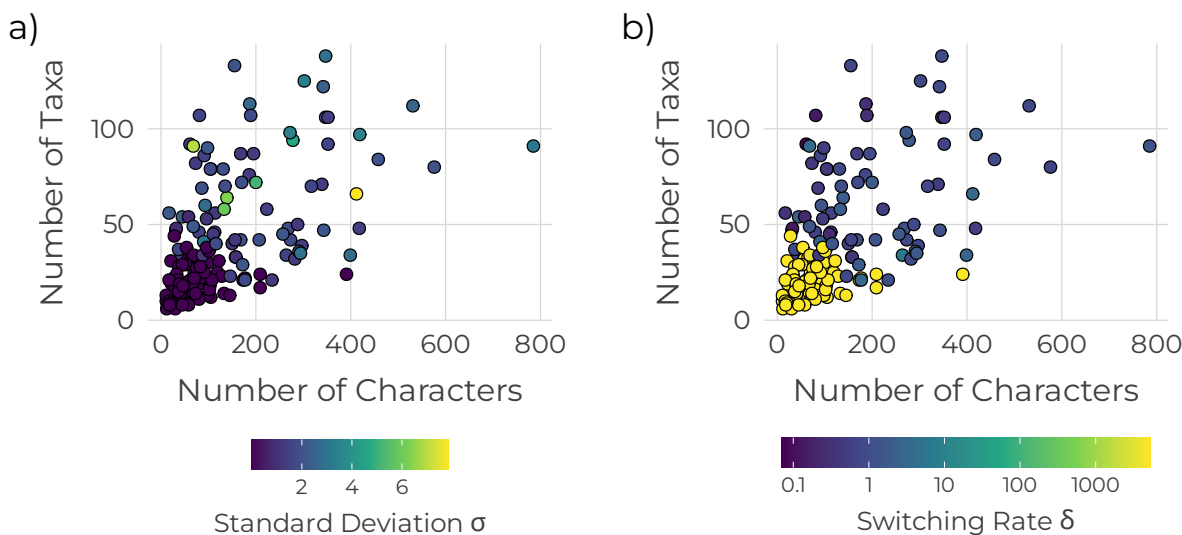


Figure S7: Plots showing the size of morphological dataset used in this study. Each point represents a dataset with corresponding number of characters and taxa colored by a) Standard deviation (σ) and b) Switching rate (δ) inferred from Covariomorph model with 4 rate categories.

Comparison between the Models

To evaluate the performance of the Covariomorph model, we compared it against the standard Mk model and the Mk model with among-character rate variation (Mk+ACRV). The comparison was conducted using model selection based on marginal likelihood estimates. For the covariomorph model, we

explored a range of different numbers of discrete rate categories (m), from 2 to 18, to assess how model fit is influenced by the number of rate categories allowed.

The marginal likelihoods for each model and for each number of rate categories were estimated for the Rays (Marramà et al., 2023) and Sharks (Shirai, 1996) datasets, as shown in Figure S8 and Table S1. Bayes factors ($\ln(BF)$) were calculated to compare the evidence for each model against the baseline Mk model. As seen in the results, the covariomorph model consistently provides a better fit to the data than both the Mk and Mk+ACRV models for both datasets, with the marginal likelihood generally increasing as more rate categories are added, before reaching a plateau. This indicates strong support for a model that allows for rates to vary across both characters and lineages.

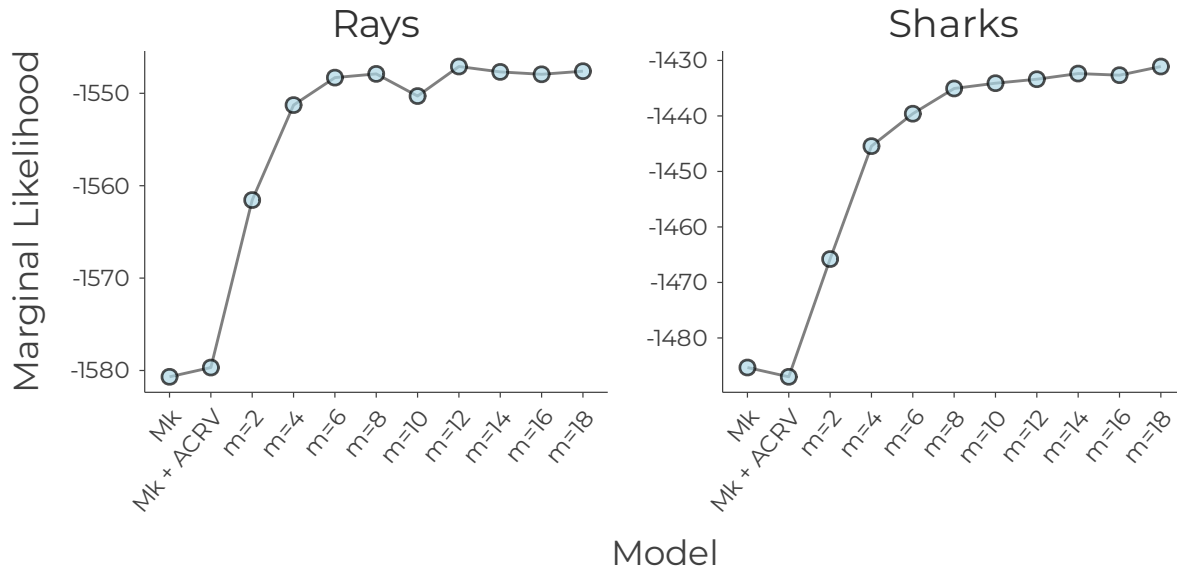


Figure S8: Marginal Likelihood estimates of Covariomorph model with varying number of rate categories

Table S1: Results of model selection analysis on Rays and Sharks dataset with varying number of covariomorph rate categories $m = \{2, 4, 6, 8, 10, 12, 14, 16, 18\}$. Bayes factor $\ln(BF)$ is calculated against the Mk model.

Dataset	Mk model $\ln(L)$	Bayes factors $\ln(BF)$ for number of rate categories (m)									
		Mk+ACRV ($m = 4$)	2	4	6	8	10	12	14	16	18
Rays	-1580.68	1.00	19.13	29.41	32.38	32.79	30.39	33.58	33.00	32.74	33.08
Sharks	-1485.32	-1.68	19.54	39.88	45.72	50.27	51.22	51.94	52.96	52.67	54.23

Topological difference

To evaluate the topological difference between the trees inferred by the different model, we calculated the normalized Robinson-Foulds (RF) score (Robinson and Foulds, 1981) between the maximum *a posteriori*

tree (MAP) (Heled and Bouckaert, 2013) from each model and the MAP tree from the Mk model for both the Rays and Sharks datasets. The RF score measures the dissimilarity between two phylogenetic trees, where a lower score indicates a higher degree of similarity.

The results in Figure S9 reveal how the choice of the model affects the topology of the inferred tree relative to the standard Mk model.

- For the Rays dataset, the RF scores were generally higher for all models, indicating a greater overall dissimilarity from the Mk model tree. Here, the Mk+ACRV model's tree showed the least dissimilarity compared to the covariomorph models.
- For the Sharks dataset, a different pattern emerged, the tree inferred using the Mk+ACRV model showed the least difference from the reference phylogeny, with the lowest RF score. The trees from the covariomorph models all exhibited greater topological distances from the Mk tree, with RF scores ranging from approximately 0.28 to 0.40.

Additionally, Figures S10 and S11 show the topological difference across the entire posterior sampled from different models. A clear trend that can be observed is that, the models without heterotachy (Mk and Mk+ACRV) are congruent with each other and the covariomorph models (with varying number of rate categories) are congruent with each other. On the other hand, the models without heterotachy and the covariomorph model are exploring different tree topologies based on the expected difference in split frequencies with a threshold of $ESS = 200$.

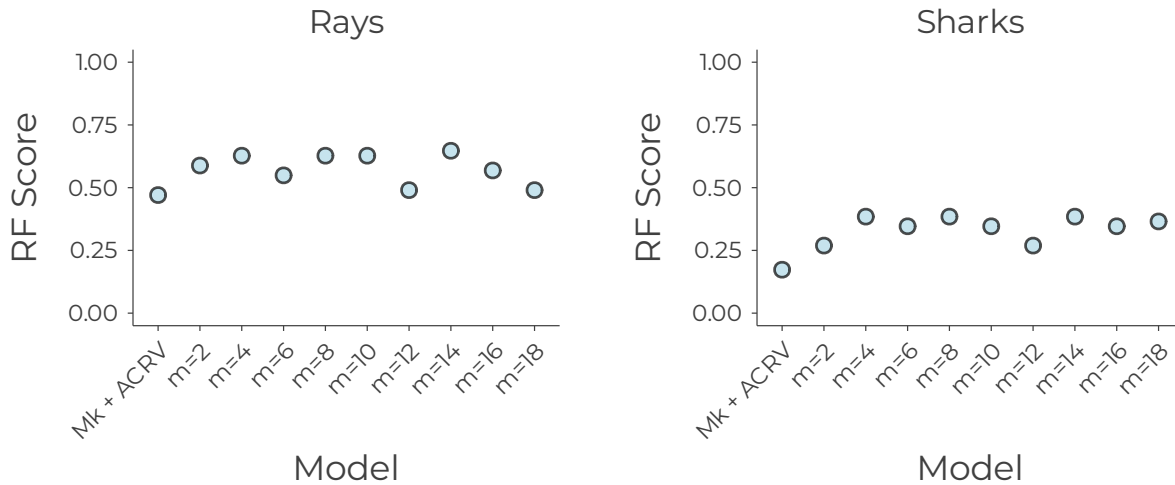


Figure S9: Normalized Robinson Foulds (RF) scores between the MAP trees for the rays and sharks datasets.

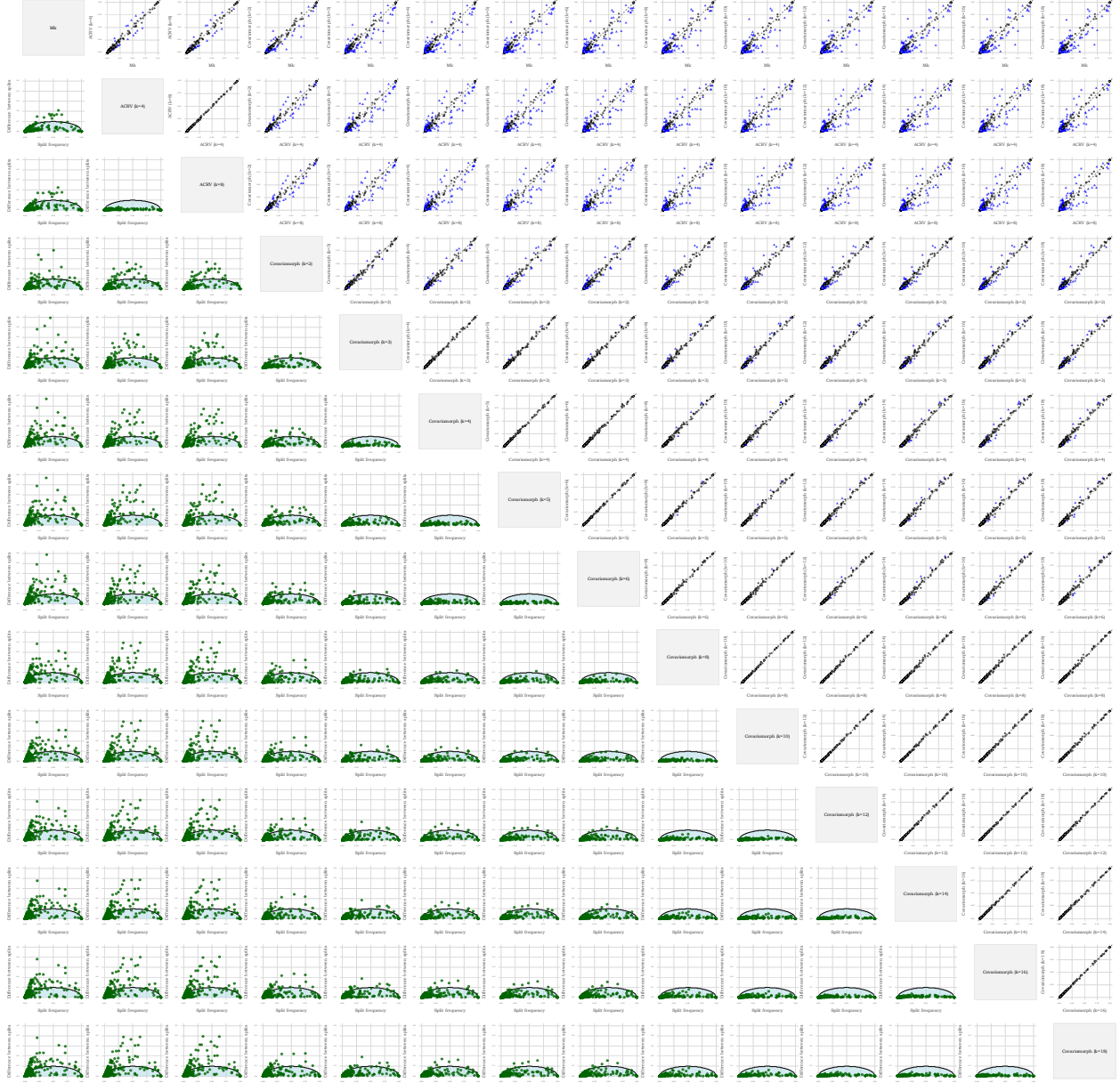


Figure S10: Differences between trees sampled by different models for the Rays dataset. The diagonal gray boxes indicate the models being compared. Each clade is represented by either a circle or a triangle in the plots. **Plots above the diagonal:** These scatter plots show the posterior probabilities of clades, illustrating the level of support each pair of models provides for the same set of potential clades. Circles indicate clades where posterior probabilities do not significantly differ under the expected difference of split frequencies test, whereas triangles represent clades with significantly different posterior probabilities. **Plots below the diagonal:** These plots display the differences in split probabilities between model pairs, with the shaded region representing the expected difference of split frequencies for an effective sample size (ESS) of 200. Clades falling within the shaded region pass the expected difference of split frequencies test. If a clade appears above the shaded semicircle, it was sampled with a significantly different probability between the two compared models.

Complete Model Comparison Matrix

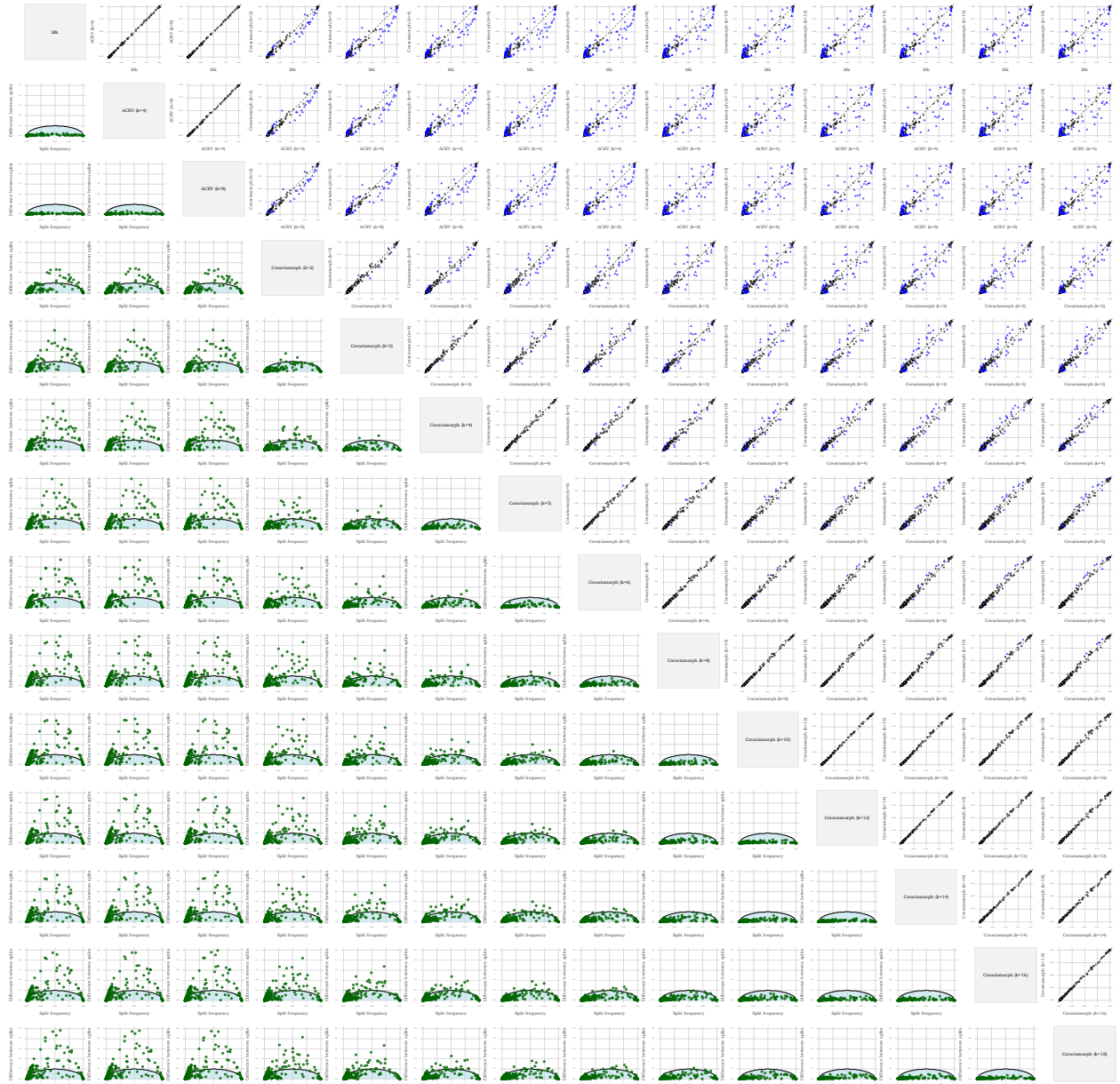


Figure S11: Differences between trees sampled by different models for the Sharks dataset. See Fig. S10 caption for more details on the interpretation.

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